

Solution Requirements

Date	26 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

This document defines the functional and non-functional requirements for the OptiCrop: Smart Agricultural Production Optimization Engine. The system uses machine learning to analyze soil nutrients and environmental conditions to recommend the most suitable crop for cultivation.

Step 1: Functional Requirements (FR)

S.NO	Requirement Category	Requirement Category	Priority
1	User Input	Allow farmers to enter Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall values.	High
2	Data Validation	Validate user input before processing.	High
3	Machine Learning Prediction	Predict the most suitable crop using the trained ML model.	High
4	Dataset Integration	Load and use the trained crop dataset and machine learning model.	High
5	Recommendation Display	Display the recommended crop clearly to the user.	High
6	Web Interface	Provide a simple and user-friendly Flask web interface. .	Medium
7	Business Rules	Generate recommendations based on soil nutrients and environmental conditions. .	High
8	Logging & Maintenance	Maintain prediction records and support future model updates.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric/ Acceptance Criteria
1	Performance & Speed	Generate crop recommendations quickly.	< 5 seconds
2	Scalability	Support multiple farmers using the application simultaneously. .	1000+ prediction requests

3	Reliability	Produce accurate and consistent crop recommendations.	High prediction accuracy
4	Availability	Ensure the application is available whenever required.	99.9% uptime
5	Usability & Accessibility	Provide a simple and intuitive user interface.	Easy to use
6	Maintainability	Allow easy updates to the machine learning model and dataset.	Modular and scalable design